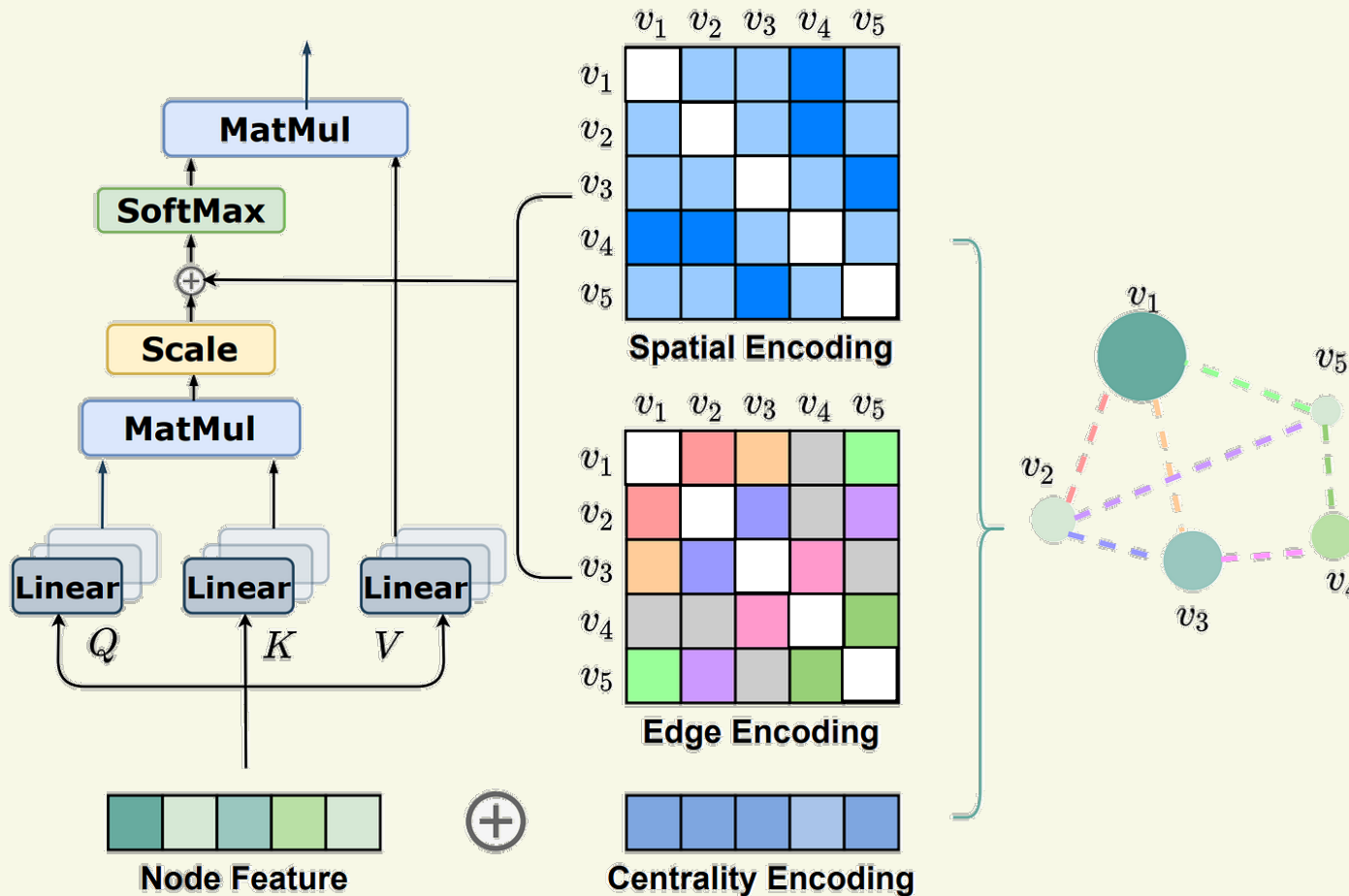


Do Transformers Really Perform Bad for Graph Representation?

Ying et al. (NeurIPS 2021)



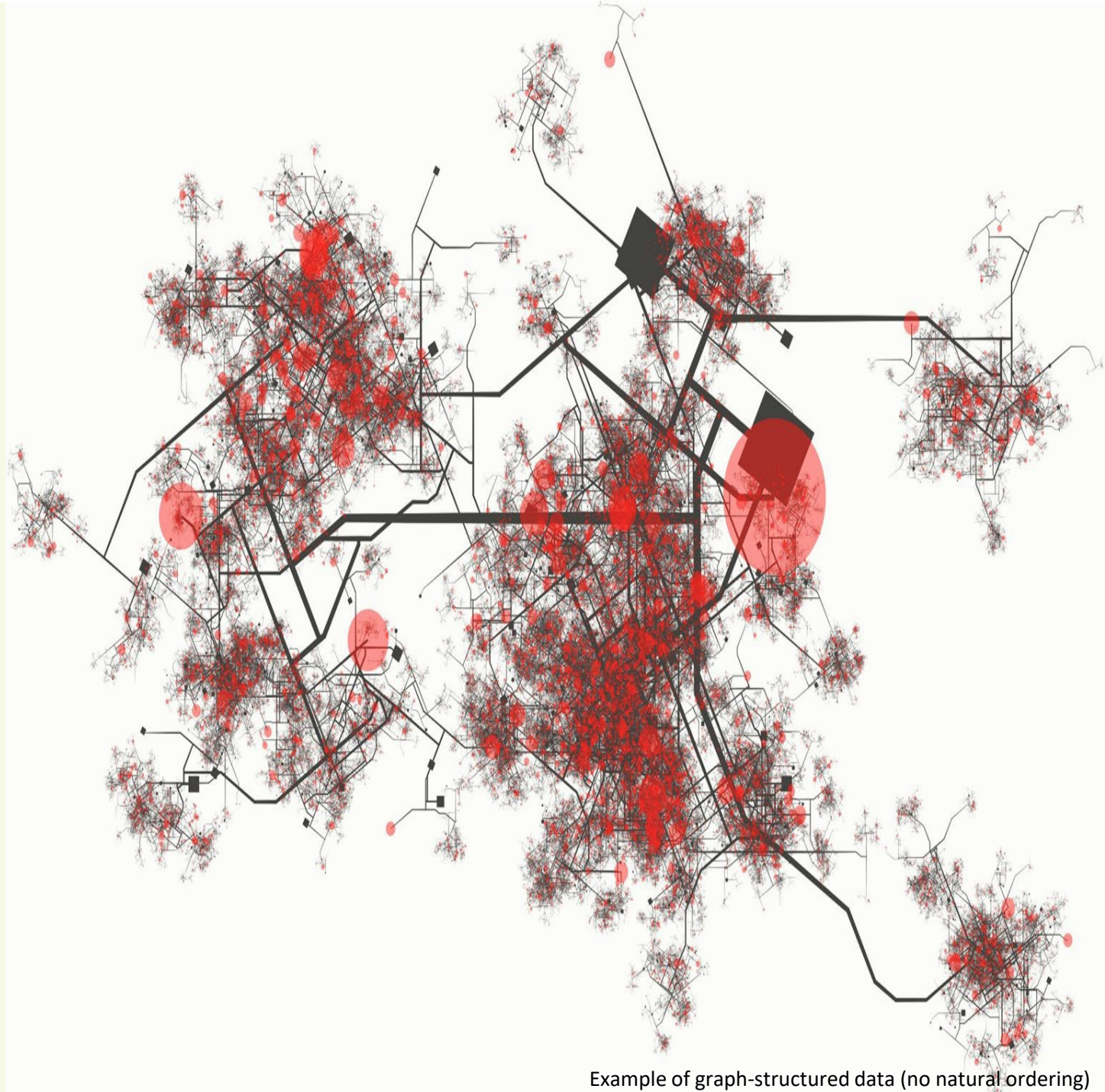
Alexander Schoolcraft
Richard Zheng
Robert Ward

Motivation:

Transformers vs Graphs

Why don't Transformers work on Graphs?

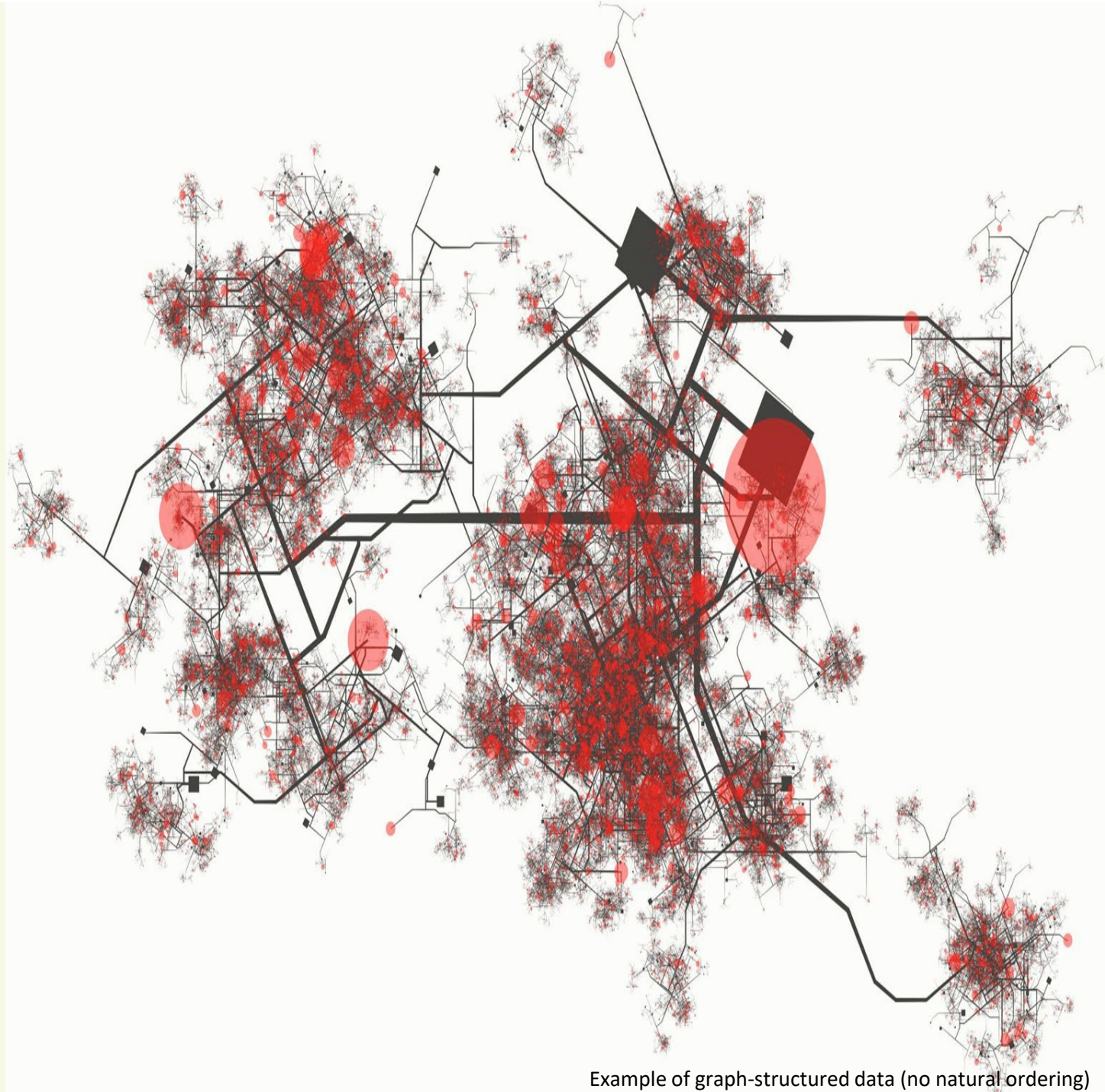
- **Transformers dominate NLP & vision**
- **But they fail to match GNNs (GCN, GIN, GraphSAGE) on graph tasks**
- **GNNs still outperform**



Example of graph-structured data (no natural ordering)

Why Graphs are hard

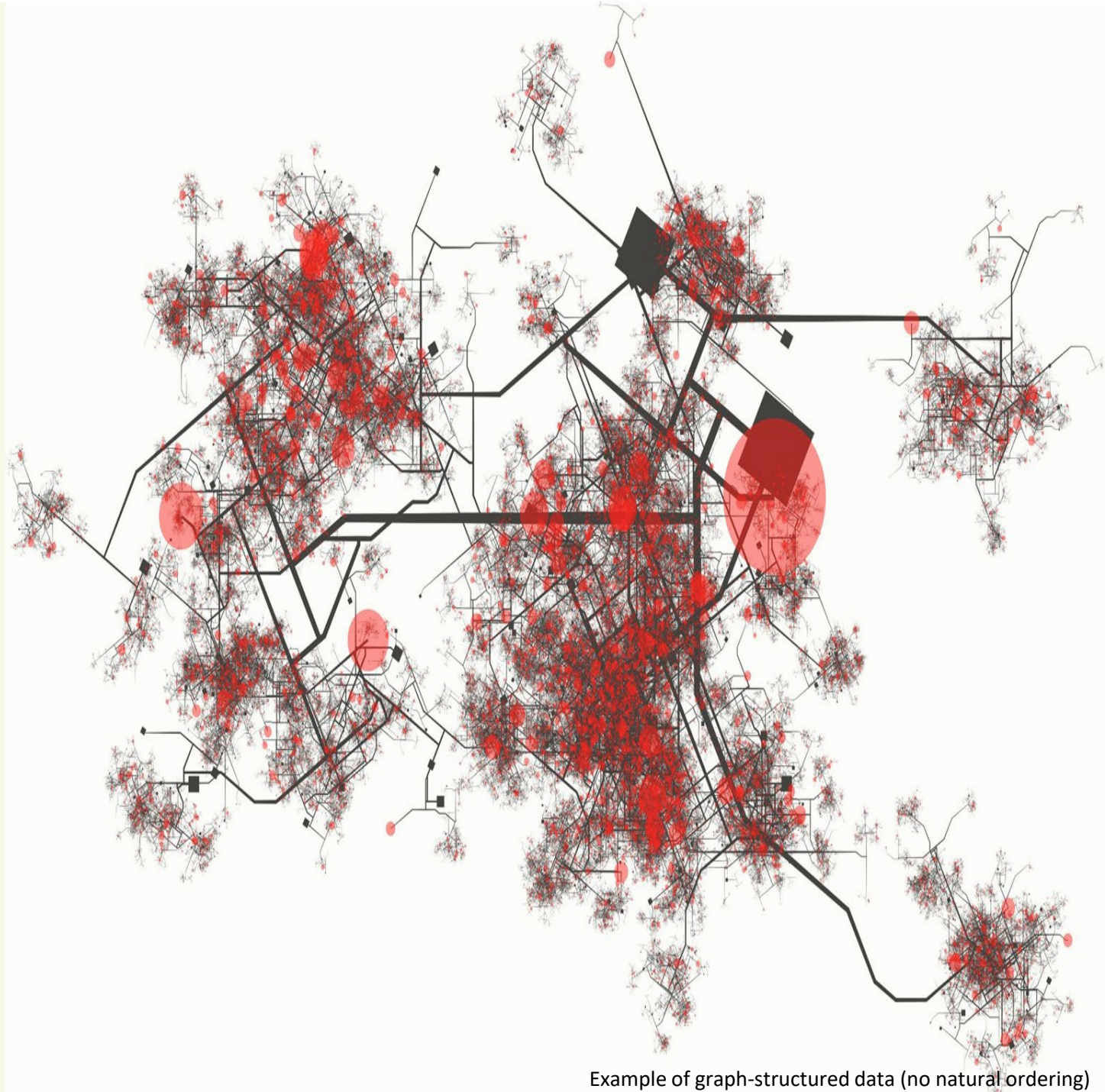
- **No natural ordering (non-Euclidean)**
- **Relationships depend on structure (connectivity, distance)**
- **Vanilla transformers ignore this structure**



Example of graph-structured data (no natural ordering)

Key Insight

- The problem is **NOT** model power
- The problem is **missing structure**
- **Solution: encode structure into attention**



Example of graph-structured data (no natural ordering)

Related Work

Graph Neural Networks (Baseline)

- **Message-passing GNNs dominate graph learning**
- **Examples: GCN, GIN, GraphSAGE**
- **Strong local structure modeling**
- **Limited global context**

Early Graph Transformers

- **Restricted Attention (local neighborhoods)**
- **Used positional/structural encodings (Laplacian, WL labeling, hop distance)**
- **Often underutilized global attention**
- **Still failed to outperform GNNs**

The Missing Piece

- **No unified way to encode graph structure**
- **Models lacked expressiveness**
- **Struggled on large benchmarks**
- **Need a unified way to encode structure**

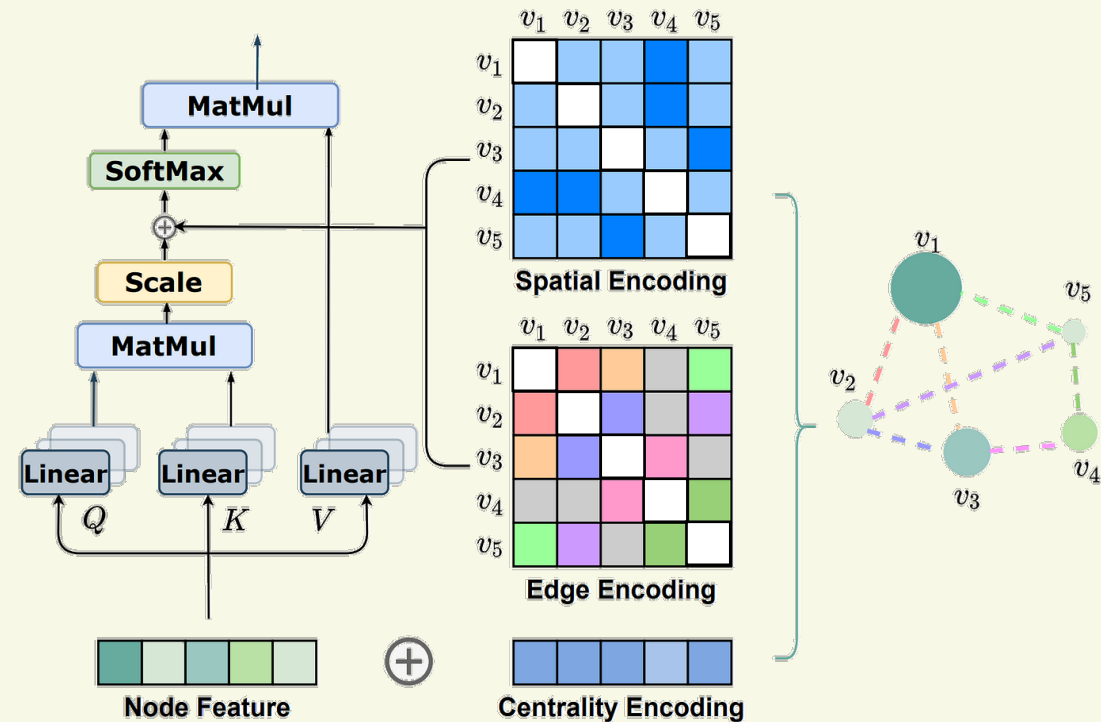
What Graphormer Adds

- **Unified way to encode graph structure**
- **Injects centrality, spatial, and edge information**
- **Unlocks full Transformer capability on graphs**

Methodology

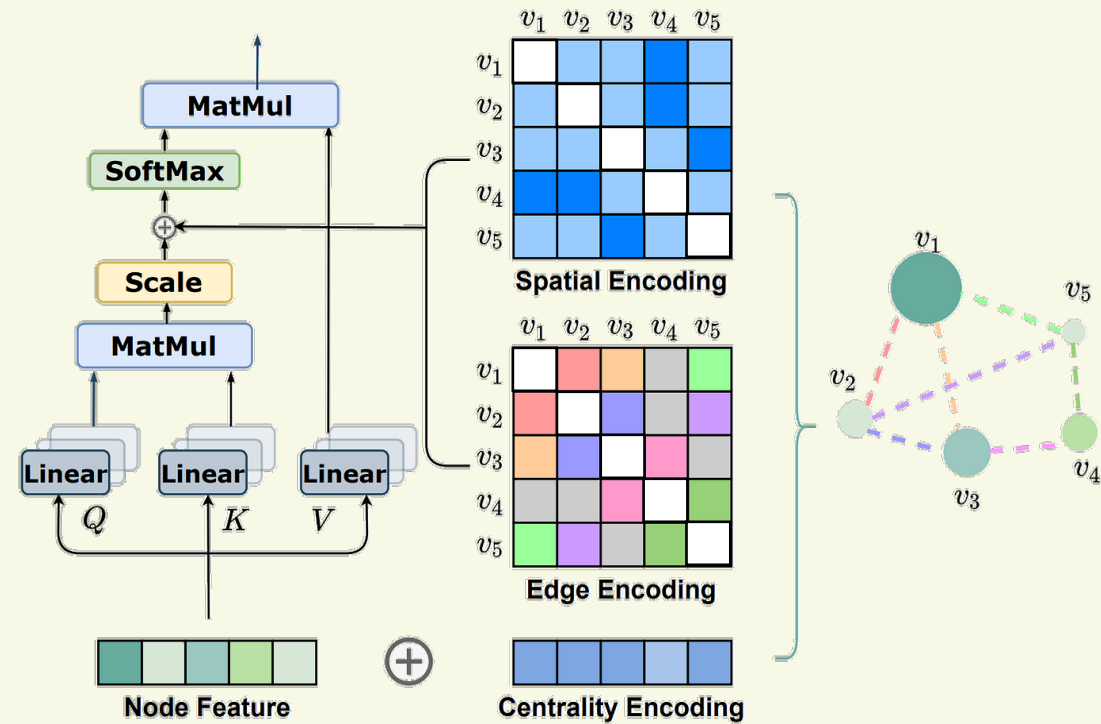
How Graphormer Works

- Standard Transformers lack graph structure
- Graphormer injects structure into attention
- Uses three key encodings:
 - Centrality
 - Spatial
 - Edge



Centrality Encoding

- Captures node importance (degree)
- Learnable embeddings per node
- Added to input features



Spatial Encoding

- **Uses shortest path distance**
- **Encoded as attention bias**
- **Captures relationships between nodes**

Edge Encoding

- **Uses edge features along paths**
- **Aggregated into attention bias**
- **Captures richer structural information**

Architecture

- **Standard Transformer encoder stack**
- **Pre-layer normalization and residual connections**
- **Virtual node for graph-level output (readout)**

Experiments & Results

Evaluation Setup

- **Benchmarked against GNNs and Transformer baselines**
- **Datasets:**
 - **OGB (PCQM4M, MolHIV, MolPCBA)**
 - **ZINC**
- **Metrics:**
 - **MAE (regression)**
 - **AUC / AP (classification)**

Key Results

- **State-of-the-art across multiple benchmarks**
- **Significant improvement on PCQM4M**
- **Outperformed GNN and Transformer baselines**
- **Consistent gains across datasets**

method	#param.	train MAE	validate MAE
GCN [26]	2.0M	0.1318	0.1691 (0.1684*)
GIN [30]	3.8M	0.1203	0.1537 (0.1536*)
GCN-VN [26, 15]	4.9M	0.1225	0.1485 (0.1510*)
GIN-VN [30, 15]	6.7M	0.1150	0.1395 (0.1396*)
GINE-VN [5, 15]	13.2M	0.1248	0.1430
DeeperGCN-VN [30, 15]	25.5M	0.1059	0.1398
GT [13]	0.6M	0.0944	0.1400
GT-Wide [13]	83.2M	0.0955	0.1408
Graphormer _{SMALL}	12.5M	0.0778	0.1264
Graphormer	47.1M	0.0582	0.1234

Why it works

- **Spatial encoding outperformed positional encodings**
- **Centrality encoding was critical**
- **Edge encoding best used as attention bias**
- **Structural encoding drives performance gains**

Critical Analysis

Strengths and Weaknesses

Strengths:

- **State-of-the-art performance**
- **Unified Transformer and GNN framework**
- **Captures both local and global structure**

Weaknesses:

- **$O(n^2)$ attention => poor scalability**
- **Requires shortest-path computation**
- **Risk of overfitting on small datasets**

Expressiveness and Application

Expressiveness:

- **Can simulate GNN message passing**
- **More expressive than 1-WL test**
- **Benefits from global attention**

Application:

- **Molecular property prediction**
- **Social network analysis**
- **Knowledge graphs**

- **Needs domain-specific encodings for best performance**

Future Work & Discussion

Future Directions

- **Reduce attention cost (scalability)**
- **Sparse or efficient Transformer variants**
- **Domain-specific structural encodings**
- **Improved graph pooling methods**

Open Questions

- **How best to encode domain knowledge?**
- **How to balance global vs local attention?**
- **How to avoid overfitting on small datasets?**

Summary

Graphormer Summary

- **Transformers fail on graphs due to missing structure**
- **Graphormer injects structure detail into attention**
- **Achieves state-of-the-art performance**
- **Opens path for future graph Transformer models**

Questions?